The Relationship Between Academic Stress and Short Term Memory Performance in Undergraduate Students

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PSY101
Summer 2026
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Submission date: 29 August 2026

Introduction

Academic stress is a very common feature of university life. Coursework, examinations, assignment deadlines, projects, time pressure and fear of poor performance can create sustained psychological pressure for undergraduate students. A review of stress in secondary and higher education concluded that academic stress can influence learning capacity, academic performance, motivation, sleep and mental health which makes it a psychologically and educationally relevant topic (Pascoe et al., 2020). This study focused on whether undergraduate students who reported different levels of academic stress also differed in immediate short term memory performance because students are often required to remember concepts, numbers, instructions and information while under pressure.

Short term memory refers to the temporary maintenance of a limited amount of information for immediate use. In the multi store model, information passes through a brief sensory register and may enter a short term store before longer term retention (Atkinson & Shiffrin, 1968). Later working memory theory distinguished temporary verbal storage from attentional control and other components (Baddeley, 2012). A forward digit span task is a widely used measure of short term verbal memory because participants must immediately reproduce sequences of digits in the same order. Compared with backward digit span, forward digit span places relatively greater emphasis on immediate storage and attention rather than manipulation of the material. Digit span performance can be represented by the number of correctly recalled trials and by the longest sequence successfully recalled (Woods et al., 2011).

Stress may influence digit span performance through changes in attention, encoding and the availability of cognitive resources. Reviews of stress biology and cognition show that stress effects vary across brain systems and exposure levels (Lupien et al., 2009). A meta analysis found that acute stress generally impaired working memory performance, although the magnitude of the effect depended on methodological and participant factors (Shields et al. 2016). These findings suggest that stress is relevant to cognitive performance but they also show that the relationship is not necessarily identical for every person or at every level of stress.

This study is guided by the Yerkes-Dodson principle. Yerkes and Dodson (1908) originally demonstrated that the relationship between simulation and learning was not always linear. The principle is now summarized as an inverted-U: very low activation may be associated with insufficient for optimal performance, a moderate level may support focused attention and task performance and very high activation may disrupt performance. This study applied this principle to academic stress and immediate memory. Importantly, the Yerkes-Dodson principle is used here as a theoretical framework rather than as a claim that every memory task must show the same curve. Therefore, the analysis tested the shape of the relationship directly rather than relying only on a simple linear correlation.

The study distinguished between academic stress over the previous two weeks and current stress immediately before the digit span task. Measuring both forms of stress allowed the project to compare broader academic pressure with the participant's immediate psychological state while linking two core topics in psychology: 1) stress and 2) memory.

This topic was selected because undergraduate students frequently experience academic stress and because memory performance is central to studying, following instructions and completing examinations. Examining stress and short term memory in the same participants provides a direct way to connect psychological theory with observable cognitive performance. The project also extends a simple stress performance question by testing both a straight line association and a possible curved relationship. This is important because a Pearson correlation alone could be zero even when performance is highest at moderate stress and lower at both ends of the stress range.

The research question was: what is the relationship between academic stress and short term memory performance in undergraduate students and is that relationship consistent with the inverted-U pattern proposed by the Yerkes-Dodson principle?

The objective of this project was to: 1) measure academic stress over the previous two weeks and their current stress immediately before a memory task, 2) assess short term memory using a forward digit span task in the same participants and 3) examine the relationship between stress and digit span performance using linear correlation and quadratic analysis in order to test for an inverted-U pattern.

Digit span performance was expected to show an inverted-U relationship with stress: performance would be relatively lower at very low and very high stress levels and relatively stronger at moderate stress levels. A negative quadratic term was therefore expected in the regression analysis.

Methodology

The study used a quantitative, cross sectional correlational design. Stress was not experimentally induced. Naturally occurring self reported stress and current stress were measured and related to performance on a forward digit span task.

The final sample contained 60 undergraduate students aged 23 to 25 years ($M = 23.95$, $SD = 0.70$). Convenience sampling was used, consistent with the approved recruitment plan involving undergraduate networks like friends and student social media groups. The sample included 38 men (63.3%) and 22 women (36.7%). Forty four participants (73.3%) were in their fourth year and sixteen (27.7%) were in their third year. Participants represented several academic

departments with the largest group coming from Computer Science & Engineering (60.0%). Participation was voluntary.

Academic stress was assessed with eight items reflecting academic demands such as examination pressure, time pressure, concern about poor performance etc. Each item was scored from 1 to 5 and the eight scores were summed to create an academic stress total score, where higher totals represented greater academic stress. These were researcher developed items, so the total score is treated as a study measure rather than as a validated clinical scale. As an exploratory reliability check, the eight items showed very high internal consistency in the present sample (Cronbach's alpha = 0.965). This value indicates that the items were strongly related to one another but it does not establish validity by itself.

Current stress was recorded immediately before the memory task using a single brief self rating of how stressed the participant felt at that moment. This measure was analysed separately from the academic stress total because it captured the participant's immediate state rather than stress over a longer period of time.

Short term memory performance was assessed with a forward digit span task. Participants were presented with increasingly long strings of digits and were required to recall the digits in the same order. The task was administered using standardized instructions and a fixed presentation rate as specified in the approved protocol. Two scores were taken: 1) the total number of correctly recalled trials and 2) highest span length passed. The highest span length passed was treated as the main outcome because it directly represents immediate forward digit span capacity.

Participants were informed of the general purpose of the study and took part voluntarily. No stress was artificially induced. Responses were recorded anonymously using participant codes rather than names or student IDs. Participants provided demographic information, academic stress item responses, a current stress rating and forward digit span responses. Data was stored for analysis without personally identifying information.

The dataset was analyzed for missing values, duplicate participant identifiers, exact duplicate rows, invalid item values and consistency of the calculated academic stress total. The dataset contained 60 complete rows and no missing values. Descriptive statistics were calculated for academic stress, current stress, highest digit span and total correct trials. Pearson correlations were used to test straight line associations between stress and memory performance. Spearman correlations were also used as a robustness check because digit span outcomes were discrete.

To test Yerkes-Dodson prediction directly, quadratic regression models were fitted for each stress indicator. Each model included a centered linear stress term and a squared stress term. A statistically significant negative squared term was interpreted as evidence of downward curvature

consistent with an inverted-U pattern. The change in R squared from the linear model to the quadratic model was examined to determine whether adding curvature improved model fit. Influential observations were screened using Cook's distance and sensitivity analyses were conducted after temporarily excluding flagged cases.

Results

Academic stress total scores ranged from 10 to 40 (M = 23.95, SD = 7.76). Current stress ratings ranged from 0 to 9 (M = 5.30, SD = 2.07). The highest forward digit span passed ranged from 4 to 9 digits (M = 6.32, SD = 1.40) while total correct digit span trials ranged from 5 to 14 (M = 10.00, SD = 2.46). Academic stress and current stress were strongly positively related, $r(58) = 0.863$, $p < 0.001$, indicating that students reporting greater academic stress over the previous two weeks also tended to report greater stress immediately before the memory task.

Table 1: descriptive statistics for main study variables

Variable	**N**	**M**	**SD**	**Min**	**Max**
Academic stress total	60	23.95	7.76	10	40
Current stress	60	5.30	2.07	0	9
Highest digit span passed	60	6.32	1.40	4	9
Correct digit span trials	60	10.00	2.46	5	14

Academic stress showed a statistically significant negative linear association with the primary memory outcome, $r(58) = -0.346$, $p = 0.007$, 95% CI [-0.551, -0.101]. Higher academic stress was therefore associated, on average, with a lower highest digit span. Academic stress was also negatively associated with total correct trials, $r(58) = -0.400$, $p = 0.002$, 95% CI [-0.594, -0.162]. Current stress showed the same general direction: it was negatively correlated with highest digit span, $r(58) = -0.427$, $p < 0.001$, 95% CI [-0.614, -0.194] and with correct trials, $r(58) = -0.390$, $p = 0.002$, 95% CI [-0.586, -0.151]. Spearman correlations produced very similar conclusions, suggesting that the linear results were not dependent on Pearson correlation alone.

The primary quadratic regression tested academic stress as a predictor of highest digit span. The quadratic model was statistically significant, $F(2, 57) = 15.38$, $p < 0.001$ and examined 35.1% of the variance in the highest span (R-squared = 0.351, adjusted R-squared = 0.328). Most importantly, the squared stress term was negative and significant, $b = -0.0106$, $SE = 0.0024$, $p <$

0.001. Adding the squared term improved explained variance by 23.1% points over the linear model, change in R-squared = 0.231, F-change(1, 57) = 20.28, $p < 0.001$. The fitted curve reached its estimated maximum at an academic stress score of approximately 22.27. So, the data were consistent with a downward curving, inverted-U pattern rather than only a straight negative line.

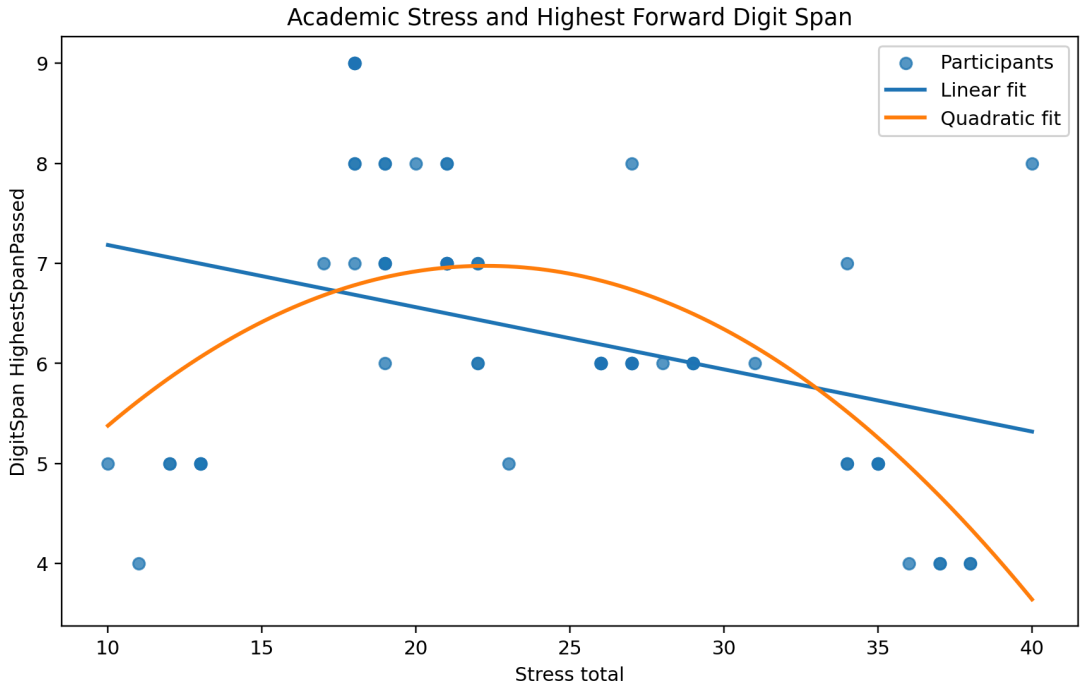


Figure 1: academic stress and highest forward digit span

The secondary analyses were consistent with the primary result. Academic stress predicting correct trials showed significant downward curvature (R-squared = 0.487, b = -0.0222, $p < 0.001$, estimated peak = 22.35). Current stress also showed significant negative quadratic terms for highest span (R-squared = 0.420, b = -0.1217, $p < 0.001$, peak = 4.10) and correct trials (R-squared = 0.499, b = -0.2584, $p < 0.001$, peak = 4.39). Therefore, the curve pattern appeared across both stress indicators and both memory outcomes.

Table 2: summary of linear and quadratic relationships

Note: $\triangle R^2$ is the increase in explained variance when the squared stress term was added to the linear model. Negative squared terms indicate downward curvature.

Predictor / outcome	Pearson r (p)	Quadratic R^2 ($\triangle R^2$)	Squared term b (p)
Academic stress / highest span	-0.346 (0.007)	0.351 (0.231)	-0.0106 (<0.001)
Academic stress / correct trials	-0.400 (0.002)	0.487 (0.327)	-0.0222 (<0.001)
Current stress / highest span	-0.427 (<0.001)	0.420 (0.238)	-0.1217 (<0.001)
Current stress / correct trials	-0.390 (0.002)	0.499 (0.347)	-0.2584 (<0.001)

Cook's distance screening identified two potentially influential observations in the primary academic stress model. When these two cases were temporarily excluded, the negative quadratic term remained statistically significant ($p < 0.001$) and the quadratic R-squared increased from 0.351 to 0.542. The estimated peak remained similar (approximately 21.73). The current stress sensitivity analysis produced the same conclusion: after excluding two flagged cases, the squared term remained significant ($p < 0.001$) and the estimated peak remained within the mid range of current stress. Thus, the main inverted-U result was not dependent on a single unusual observation.

Discussion

The purpose of this study was to examine the relationship between academic stress and short term memory performance in undergraduate students and to test whether the relationship was consistent with an inverted-U pattern. The results supported two related conclusions: 1) when the relationship was treated as a simple straight line, greater stress was associated with poorer digit span performance and 2) more importantly for hypothesis, the quadratic models fit the data substantially better than the linear model and showed significant negative curvature. Memory performance was relatively strongest around moderate stress levels and weaker toward the lower and higher ends of the observed stress ranges. This pattern appeared for both academic stress and current stress and for both digit span outcomes.

The primary analysis was consistent with the Yekes-Dodson framework. The fitted academic stress curve peaked around a total score of 22.27, closed to the middle of the observed 10-40 range and the current stress curve peaked around 4.10. These values should not be interpreted as universal or clinically meaningful cut off scores. The academic stress items were researcher developed and the estimated peaks are sample specific points on fitted regression curve. However, the overall shape of the relationship supports the idea that cognitive performance may depend on the level of stress rather than changing in a perfectly linear way. Moderate stress may

help maintain task management and attention whereas stronger stress may consume attentional resources or interfere with the temporary maintenance of information.

The negative linear correlations are also psychologically meaningful. Across the sample as a whole, higher academic stress and higher current stress were associated with lower digit span performance. This is broadly consistent with research showing that stress can disrupt working memory and executive control. Shields et al. (2016) reported in a meta analysis that acute stress generally impaired working memory.

The findings also fit research on student stress. Pascoe et al. (2020) reviewed evidence that academic stress can interfere with learning and academic performance. This study did not measure perceived control or physiological stress responses but its curved pattern is compatible with the broader idea that stress effects vary with task demands and individual differences.

The study also distinguished short term memory from more complex working memory manipulation. The forward digit span task primarily required participants to maintain and immediately reproduce verbal information in sequence. This matches the project's focus on short term memory and the idea of a limited capacity temporary store described in classic memory theories (Atkinson & Shiffrin, 1968). Modern working memory theory also emphasizes the role of attention and temporary verbal storage (Baddeley, 2012). Under high stress, attention may be diverted toward worries, performance concerns, time pressure or other stress related thoughts, leaving fewer resources available for encoding and retaining the digits. This explanation is plausible but the present correlational design cannot directly identify the cognitive or biological mechanism responsible for the observed pattern.

A strength of the findings is their consistency across measures. Highest span and total correct trials both produced significant negative quadratic terms, academic stress and current stress produced similar curves. Sensitivity analyses also showed that the quadratic conclusion remained after temporarily removing observations flagged by Cook's distance. This convergence reduces the likelihood that the main result was produced by one scoring method or one unusual participant.

At the same time, the Yerkes-Dodson interpretation should remain cautious. The original Yerkes-Dodson experiment (1908) examined stimulus strength and habit learning under specific task conditions whereas this project measured naturally occurring self reported stress and a digit span task. Therefore, this study does not prove a universal law of optimal stress. A more appropriate conclusion is that the observed relationship was statistically consistent with an inverted-U pattern in this sample. The result is useful because the analysis directly tested curvature instead of assuming that a zero or negative correlation alone could answer the theoretical question.

Limitations

Several limitations should be considered. Firstly, the sample was relatively small and was selected using convenience sampling. The participants were also concentrated in a narrow age range, most were fourth year students and 60% were from CSE These features limit how confidently the findings can be generalized to all undergraduate students. Secondly, academic stress was assessed using researcher developed items rather than a previously validated academic stress scale. Though internal consistency was high, reliability is not the same as construct validity and some items may have been redundant. The current stress measure was also a single self report item.

Thirdly, the study was correlational and cross sectional so it cannot establish that stress caused differences in digit span performance. Unmeasured factors such as sleep, caffeine, time of day, recent studying etc may have influenced both stress and memory. Fourth, the fitted peak estimates are sample specific and should not be treated as ideal stress targets. A larger, more controlled study would be needed to estimate the shape of the relationship more precisely.

Conclusion and Recommendations

This study investigated whether academic stress was related to short term memory performance in 60 undergraduate students. Higher stress was associated with poorer digit performance in the linear analyses. However, the main finding was that quadratic models provided a substantially better description of the data: both academic stress and current stress showed significant downward curving relationships with highest digit span and total correct trials. The pattern was consistent with an inverted-U interpretation in which moderate stress was associated with relatively stronger memory performance and higher stress was associated with poorer performance. Sensitivity analyses suggested that this result was not dependent on only one or two influential observations.

The result supports practical attention to academic stress, especially during periods of heavy workload. Students may benefit from planning coursework early, avoiding unnecessary deadline accumulation, using regular breaks and maintaining sleep and study routines that reduce excessive stress before cognitively demanding tasks. Universities can also support students through realistic scheduling, academic advising and accessible stress management resources. These recommendations should be understood as ways to reduce excessive academic pressure rather than as an attempt to create a particular optimal stress score.

Future study should use a larger and more diverse sample including students from multiple institutions and years of study and use a validated academic stress measure alongside the current stress rating. Repeated measurements across low and high pressure would help separate stable

individual differences from changes within the same student. Researchers could also record sleep, caffeine intake, time of day and prior task experience and could use additional short term or working memory tasks. Such improvements would provide a stronger test whether the curved pattern found in this project is reliable and generalizable.

References

Pascoe, M. C., Hetrick, S. E., & Parker, A. G. (2020). The impact of stress on students in secondary school and higher education. *International Journal of Adolescence and Youth, 25*(1), 104-112. https://doi.org/10.1080/02673843.2019.1596823

Atkinson, R. C., & Shiffrin, R. M. (1968). Human memory: A proposed system and its control processes. In K. W. Spence & J. T. Spence (Eds.), *The psychology of learning and motivation: Advances in research and theory* (Vol. 2, pp. 89-195). Academic Press. https://doi.org/10.1016/S0079-7421(08)60422-3

Baddeley, A. (2012). Working memory: Theories, models, and controversies. *Annual Review of Psychology, 63*, 1-29. https://doi.org/10.1146/annurev-psych-120710-100422

Woods, D. L., Kishiyama, M. M., Yund, E. W., Herron, T. J., Edwards, B., Poliva, O., Hink, R. F., & Reed, B. (2011). Improving digit span assessment of short term verbal memory. *Journal of Clinical and Experimental Neuropsychology, 33*(1), 101-111. https://doi.org/10.1080/13803395.2010.493149

Lupien, S.J., McEwen, B. S., Gunnar, M. R., & Heim, C. (2009). Effects of stress throughout the lifespan on the brain, behaviour and cognition. *Nature Reviews Neuroscience, 10*(6), 434-445. https://doi.org/10.1038/nrn2639

Shields, G. S., Sazma, M. A., & Yonelinas, A. P. (2016). The effects of acute stress on core executive functions: A meta analysis and comparison with cortisol. *Neuroscience & Biobehavioral Reviews, 68*, 651-668. https://doi.org/10.1016/j.neubiorev.2016.06.038

Yerkes, R. M., & Dodson, J. D. (1908). The relation of strength of stimulus to rapidity of habit formation. *Journal of Comparative Neurology & Psychology, 18*(5), 469-482. https://doi.org/10.1002/cne.920180503

Appendix

Table A1: study variable and scoring summary

Variable	**Meaning / scoring**
Academic stress total	Sum of eight researcher developed items scored 1-5, higher total = greater academic stress
Current stress	Single immediate self rating 0-10 obtained directly before the memory task
Highest digit span passed	Longest forward digit sequence successfully recalled in the correct order, primary memory outcome
Correct digit span trials	Total number of correctly recalled trials across the forward digit span task, supporting memory outcome